

The "For Dummies" Guide: IoT Biometric Attendance System

A Complete Manual for Operation, Hardware, and Future Scalability

1. What is this Project? (The "Non-Technical" View)

Imagine a digital sign-in sheet that can't be faked. This device uses your unique fingerprint to mark your attendance.

- **The Goal:** To stop "proxy" attendance (where one friend signs for another).
 - **The Result:** A fast, beep-confirmed system that shows your Name, Roll No, and Branch on a screen the moment you touch it.
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2. Why these "Parts" were chosen? (The Technical Logic)

The Brain: ESP32 Microcontroller

We didn't use a standard Arduino because the **ESP32** is like a smartphone processor compared to a calculator.

- **Dual-Core:** One "brain" handles the fingerprint sensor while the other handles the screen and memory.
- **Built-in Memory:** It has "Internal Flash" memory, so it remembers names even when the power is cut.
- **WiFi-Ready:** It has built-in internet capability for future cloud updates.

The Eye: R307 Optical Sensor

This is a high-speed camera that "sees" the ridges of your finger.

- **Fast Search:** It can search through 1,000 fingers in less than a second.
- **On-Board Processing:** It doesn't just take a picture; it converts your finger into a mathematical code (a "template") so your actual fingerprint image is never stored, protecting your privacy.

The Interface: I2C LCD & Buzzer

- **LCD:** We used an **I2C version** because it only needs 4 wires instead of 16, making the project neat and less likely to have loose connections.
 - **Buzzer:** We chose an **Active Buzzer** to give "Audio Cues"—like a car's lock sound—so you know if you've succeeded without looking at the screen.
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3. Hardware Connections (The "Map")

Component	Pin on ESP32	Why this Pin?
Fingerprint (TX)	GPIO 16	Part of "Hardware Serial 2" for high-speed, lag-free data.
Fingerprint (RX)	GPIO 17	Dedicated communication line for commands.
LCD (SDA)	GPIO 21	Standard Data line for the I2C communication protocol.
LCD (SCL)	GPIO 22	Standard Clock line to keep the screen data in sync.
Buzzer (+)	GPIO 15	A general-purpose pin that doesn't interfere with the ESP32's "startup" sequence.

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4. Where is the Information Stored?

One of the most common questions is: *"Where do the names go?"*

1. **Fingerprints:** Stored inside the **R307 Sensor's** own internal flash chip as "Template IDs" (1, 2, 3...).
 2. **Student Details:** Stored in the **ESP32's NVS (Non-Volatile Storage)** using the Preferences library.
 - Think of it like a small built-in SD card.
 - We link them like this: **ID #1 (Sensor) = "Omkar | CS | 45" (ESP32 Memory)**.
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5. How to Operate (The User Manual)

Registering a New Person:

1. Connect the device to your PC and open the **Serial Monitor** (115200 Baud).
2. Type 'r' and press Enter.
3. **The 3-Scan Rule:** Place your finger when asked. You must do this **3 times**.
 - *Tip:* Shift your finger slightly on the 3rd scan to capture the edges for better accuracy.

4. Type the **Name, Branch, and Roll No** when prompted in the Serial bar.

Taking Attendance:

1. Simply place a registered finger on the sensor.
 2. **Green Light/Long Beep:** Attendance Marked! Your name appears on the LCD.
 3. **Red Light/Double Beep:** Unrecognized finger. Try again.
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6. Security & Future Scope

Making it "More Secure"

- **Encryption:** Currently, data is stored in plain text. A more secure version would "scramble" (encrypt) the names in the memory so no one can steal the student list.
- **Tamper Switch:** Add a small button inside the box. If someone tries to open the device, it wipes the memory automatically.

Future Scope (Version 2.0)

- **Web Dashboard:** Since the ESP32 has WiFi, we can send this data to a website so parents get an SMS the moment their child enters the college.
 - **Face Recognition:** Adding a small ESP32-CAM to double-check the face along with the fingerprint (Multi-factor Authentication).
 - **SD Card Logging:** Adding a physical SD card slot so the device can store 10 years' worth of attendance data offline.
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Quick Troubleshooting for Dummies:

- **Screen is blank?** Adjust the small "Blue Potentiometer" on the back of the LCD with a screwdriver to fix the contrast.
- **Not detecting?** Wipe the sensor glass with a clean cloth. Oils from fingers can "blur" the lens.
- **No Beep?** Check if the Buzzer is plugged in correctly—the long leg must be in Pin 15.